



PAPER – 2

ADVANCED FINANCIAL MANAGEMENT



QUESTIONS

Risk Management

1. Mr. B is a rational risk taker. He takes his position in a single stock for 4 days in a week. He does not take a position on Friday to avoid weekend effect and takes position only for four days in a week i.e. Monday to Thursday. He transfers the amount on Monday morning and withdraws the balance on Friday morning. He desires to take a maximum exposure in the single stock (not the portfolio) where Value at Risk (VAR) should not exceed the balance lying in his bank account. The position by his manager, as per standing instructions, is taken on the free balance lying in the bank account in the morning on each Monday.

On Monday morning (before opening of the capital market) he has transferred an amount of ₹ 11 Crore to his bank account. A fixed deposit also matured on this Monday. The maturity amount of ₹ 63,42,560 was also credited to his account by the bank in the morning of the Monday. However, Mr. B received the intimation of the same in the evening. The bank needs a minimum balance of ₹ 1,000 all the time.

The other information with respect to stocks X and Y, which are under consideration for this week, is as under:

X		Y	
Return	Probability	Return	Probability
6	0.10	4	0.10
7	0.25	6	0.20
8	0.30	8	0.40

9	0.25	10	0.20
10	0.10	12	0.10

Based on above information, answer the following questions:

- I. Available amount which can be used by Mr. B for potential exposure for 4 days on Monday morning shall be.....
- (a) ₹ 11,00,00,000
(b) ₹ 11,63,41,560
(c) ₹ 11,00,01,000
(d) ₹ 11,63,42,560
- II. The Z-score at a 99% confidence level for Mr. B's Value at Risk (VaR) is
- (a) 1.64
(b) 1.96
(c) 2.33
(d) 2.58
- III. The expected return for the stocks X is
- (a) 7%
(b) 8%
(c) 9%
(d) 10%
- IV. The expected return for the stocks Y is
- (a) 7%
(b) 8%
(c) 9%
(d) 10%

- V. In which stock should Mr. B invest in to maximize his returns while maintaining his Value at Risk (VaR) within acceptable limits?
- (a) Stock X
 - (b) Stock Y
 - (c) Both stocks are equally good
 - (d) Neither stock is suitable

Interest Rate Risk Management

2. P Ltd. is planning to borrow an amount of ₹ 60 crores for a period of 3 months in the coming 6 month's time from now. The current rate of interest is 9% p.a., but it is likely to go up in 6 month's time. The company wants to hedge itself against the likely increase in interest rate.

You as CFO has been asked to suggest both traditional as well as modern methods to hedge interest rate risk.

Suppose the banker of P Ltd. has quoted the following Forward Rate Agreement (FRA) rates:

3 x 6	9.10%	9.15%
6 x 9	9.20%	9.30%
9 x 12	9.35%	9.45%

Based on the above information answer the following questions:

- I. Suppose if P Ltd. agrees to adopt FRA method to hedge interest rate risk then the interest rate..... shall be applicable for the same agreement.
 - (a) 9.10% p.a.
 - (b) 9.30% p.a.
 - (c) 9.35% p.a.
 - (d) 9.45% p.a.
- II. Suppose if the actual rate of interest after 6 months happens to be 9.60%, then the settlement amount approximately
 - (a) ₹ 733,855 shall be paid by P Ltd. to its Banker.

- (b) ₹ 439,453 shall be paid by P Ltd. to its Banker.
 - (c) ₹ 439,453 shall be paid by Banker to P Ltd.
 - (d) ₹ 733,855 shall be paid by Banker to P Ltd.
- III. Suppose if the actual rate of interest after 6 months happens to be 8.80%, then the settlement amount approximately
- (a) ₹ 733,855 shall be paid by P Ltd. to its Banker
 - (b) ₹ 439,453 shall be paid by P Ltd. to its Banker.
 - (c) ₹ 439,453 shall be paid by Banker to P Ltd.
 - (d) ₹ 733,855 shall be paid by Banker to P Ltd.
- IV. Which of the following technique is not the modern technique to hedge the interest rate risk
- (a) Interest Rate Futures
 - (b) Interest Rate Options
 - (c) Interest Rate Swaps
 - (d) Forward Rate Agreement

Derivatives Analysis and Valuation

3. Mr. X, is a Senior Portfolio Manager at ABC Asset Management Company. He expects to purchase a portfolio of shares in 90 days. However, he is worried about the expected price increase in shares in coming day and to hedge against this potential price increase he decides to take a position on a 90-day forward contract on the Index. The index is currently trading at 2290. Assuming that the continuously compounded dividend yield on the same index is 1.75% and risk-free rate of interest is 4.16%, you are required to determine:
- (a) The justified forward price on this contract.
 - (b) The position Mr. X should take in forward contract on the Index.
 - (c) Gain/ loss on the position taken if after 28 days of the purchase of the contract the Index value stands at 2450.
 - (d) Gain/ loss on the position taken if at expiration of 90 days the Index Value is 2470.

Note: Take 365 days in a year and value of $e^{0.005942} = 1.005960$, $e^{0.001849} = 1.001851$.

4. AB Ltd.'s equity shares are presently selling at a price of ₹ 500 each. An investor is interested in purchasing AB Ltd.'s shares. The investor expects that there is a 70% chance that the price will go up to ₹ 650 or a 30% chance that it will go down to ₹ 450, three months from now. There is a call option on the shares of the firm that can be exercised only at the end of three months at an exercise price of ₹ 550.

Calculate the following:

- (i) If the investor wants a perfect hedge, what combination of the share and option should he select?
- (ii) Explain how the investor will be able to maintain identical position regardless of the share price.
- (iii) If the risk-free rate of return is 20% p.a. for the three months period, what is the value of the option at the beginning of the period?
- (iv) What is the expected rate of return for the option buyer?

Business Valuation

5. ABC, a large business house is planning to sell its wholly owned subsidiary KLM. Another large business entity XYZ has expressed its interest in making a bid for KLM. XYZ expects that after acquisition the annual earning of KLM will increase by 10%.

Following information, ignoring any potential synergistic benefits arising out of possible acquisitions, are available:

- (i) Profit after tax for KLM for the financial year which has just ended is estimated to be ₹ 10 crore.
- (ii) KLM's after-tax profit has an increasing trend of 7% each year and the same is expected to continue.
- (iii) Estimated post tax market return is 10% and risk-free rate is 4%. These rates are expected to continue.
- (iv) Corporate tax rate is 30%.

	XYZ	ABC	Proxy entity for KLM in the same line of business
No. of shares	100 lakhs	80 lakhs	--
Current share price	₹ 287	₹ 375	--
Dividend pay out	40%	50%	50%
Debt: Equity at market values	1 : 2	1 : 3	1 : 4
P/E ratio	10	13	12
Equity beta	1	1.1	1.1

Assume that gearing level of KLM to be the same as for ABC and a debt beta is zero.

You are required :

- To calculate appropriate cost of equity for KLM based on the data available for the proxy entity.
- A range of values for KLM both before and after any potential synergistic benefits to XYZ of the acquisition.
- Compute the market value of KLM as a part of ABC.

Note: Round off calculation up to 2 decimal and compute figure in ₹ crores.

Foreign Exchange Exposure and Risk Management

- M/s Omega Electronics Ltd. exports air conditioners to Germany by importing all the components from Singapore. The company is exporting 2,400 units at a price of Euro 500 per unit. The cost of imported components is S\$ 800 per unit. The fixed cost and other variables cost per unit are ₹ 1,000 and ₹ 1,500 respectively. The cash flows in Foreign currencies are due in six months. The current exchange rates are as follows:

₹/Euro	51.50/55
₹/S\$	27.20/25

After six months the exchange rates turn out as follows:

₹/Euro	52.00/05
₹/\$	27.70/75

- (A) You are required to calculate loss/gain due to transaction exposure.
- (B) Based on the following additional information calculate the loss/gain due to transaction and operating exposure if the contracted price of air conditioners is ₹ 25,000 :

- (i) the current exchange rate are as follows:

₹/Euro	51.75/80
₹/\$	27.10/15

- (ii) Price elasticity of demand is estimated to be 1.5

- (iii) Payments and receipts are to be settled at the end of six months.

7. Z Ltd. importing goods worth USD 2 million, requires 90 days to make the payment. The overseas supplier has offered a 60 days interest free credit period and for additional credit for 30 days an interest of 8% per annum.

The bankers of Z Ltd offer a 30 days loan at 10% per annum and their quote for foreign exchange is as follows:

	₹
Spot 1 USD	82.50
60 days forward for 1 USD	83.40
90 days forward for 1 USD	83.90

You are required to evaluate the following options:

- (i) Pay the supplier in 60 days, or
- (ii) Avail the supplier's offer of 90 days credit.

Note: -

1. Consider 360 day a year.

2. Round off calculation upto whole number except the applicable rates.

Advanced Capital Budgeting Decisions

8. A firm has an investment proposal, requiring an outlay of ₹ 12 crore. The investment proposal is expected to have two years economic life with no salvage value. In year 1, there is a 0.7 probability that cash inflow after tax will be ₹ 7.50 crore and 0.3 probability that cash inflow after tax will be ₹ 9 crore. The probability assigned to cash inflow (after tax) for the year 2 corresponding to the year / Cash Inflow is as follows:

(₹ in crore)

The cash inflow year 1	₹ 7.50	₹ 9.00
The cash inflow year 2	Probability	Probability
	₹ 3.60 0.10	₹ 6.00 0.50
	₹ 4.80 0.50	₹ 7.50 0.30
	₹ 6.60 0.40	₹ 9.00 0.20

The firm uses a 15% discount rate for this type of investment.

Required:

- Construct a decision tree for the proposed investment project.
- Calculate the expected net present value (NPV).
- What will be the best outcome and the probability of that occurrence?
- What net present value will the project yield if worst outcome is realized and What is the probability of occurrence of this NPV?
- Advice whether the project be accepted or not.

Note:

- 15% discount factor 1 year 0.870; 2 year 0.756.
- Carryout all calculations in ₹ crore & round off them upto 2 decimal points.

Portfolio Management

9. Mr. Hari Kumar has categorized all the available stock in the market into the following types:

- (i) Small cap growth stocks
- (ii) Small cap value stocks
- (iii) Large cap growth stocks
- (iv) Large cap value stocks

Mr. Hari Kumar also estimated the weights of the above categories of stocks in the market index. Further, the sensitivity of returns on these categories of stocks to the three important factor are estimated to be as follows:

Category of Stocks	Weight in the Market Index	Factor I (Beta)	Factor II (Book Price)	Factor III (Inflation)
Small cap growth	15%	0.65	1.95	1.65
Small cap value	20%	1.25	2.23	2.15
Large cap growth	15%	2.25	3.20	8.65
Large cap value	50%	1.325	2.25	9.50
Risk Premium		8.85%	-4.25%	0.80%

The rate of return on treasury bonds is 5.80%

Required:

- (a) Using Arbitrage Pricing Theory, determine the expected return on the market index.
- (b) Using Capital Asset Pricing Model (CAPM), determine the expected return on the market index.
- (c) Mr. Hari Kumar wants to construct a portfolio constituting only the 'small cap growth' and 'large cap value' stocks. If the target beta for the desired portfolio is 1, suggest what should be the composition of the same.

Note: Round off calculations upto 3 decimal points.

Security Valuation

10. Mr. Z will need ₹ 18,00,000 after two years for which he wants to make one-time necessary investment now. He has a choice of two types of bonds. Their details are as below:

	Bond X	Bond Y
Face value	₹ 1,000	₹ 1,000
Coupon	8% payable annually	9% payable annually
Years to maturity	1	4
Current price	₹ 964.44	₹ 909.42
Current yield	12%	12%

Advice Mr. Z whether he should invest all his money in one type of bond or he should buy both the bonds and, if so, in which quantity? Assume that there will not be any call risk or default risk.

Note: -

1. Use PVFs upto 3 decimal points.
 2. While the numbers of bonds to be acquired to be rounded off, the remaining calculation shall be rounded off upto 3 decimal points.
11. Capital structure of Sun Ltd., as at 31.3.2023 was as under:

	(₹ in lakhs)
Equity share capital (₹ 100 each)	600
6% Preference share capital	300
9% Debentures	480
Reserves	240

Sun Ltd., earns a profit of ₹ 240 lakhs annually on an average before deduction of income-tax, which works out to 30%, and interest on debentures.

Normal return on equity shares of companies similarly placed is 7.2% provided:

- (a) Profit after tax covers fixed interest and fixed dividends at least 3 times.*
- (b) Capital gearing ratio is 0.5625.
- (c) Yield on equity shares capital is calculated at 50% of profits distributed and at 5% on undistributed profits.

Sun Ltd., has been regularly paying equity dividend of 6%.

Compute the value per equity share of the company assuming that adjustment for risk premium in comparison of similarly placed company shall be made as follows:

- (i) 1% for every one time of difference for Interest and Fixed Dividend Coverage.
- (ii) 2% for every one time of difference for Capital Gearing Ratio.

$$\frac{\text{PAT} + \text{Debenture Interest}}{\text{Debenture Interest} + \text{Preference Dividend}}$$

* $\frac{\text{PAT} + \text{Debenture Interest}}{\text{Debenture Interest} + \text{Preference Dividend}}$

Mergers, Acquisitions and Corporate Restructuring

12. R Ltd. and S Ltd. are companies that operate in the same industry. The financial statements of both the companies for the current financial year are as follows:

Balance Sheet

Particulars	R. Ltd. (₹)	S. Ltd (₹)
Equity & Liabilities		
Shareholders Fund		
Equity Capital (₹ 10 each)	20,00,000	16,00,000
Retained earnings	4,00,000	-
Non-current Liabilities		
16% Long term Debt	10,00,000	6,00,000
Current Liabilities	<u>14,00,000</u>	<u>8,00,000</u>
Total	<u>48,00,000</u>	<u>30,00,000</u>

Assets		
Non-current Assets	20,00,000	10,00,000
Current Assets	<u>28,00,000</u>	<u>20,00,000</u>
Total	<u>48,00,000</u>	<u>30,00,000</u>

Income Statement

	Particulars	R. Ltd. (₹)	S. Ltd. (₹)
A.	Net Sales	69,00,000	34,00,000
B.	Cost of Goods sold	<u>55,20,000</u>	<u>27,20,000</u>
C.	Gross Profit (A-B)	13,80,000	6,80,00
D.	Operating Expenses	4,00,000	2,00,000
E.	Interest	<u>1,60,000</u>	<u>96,000</u>
F.	Earnings before taxes [C-(D+E)]	8,20,000	3,84,000
G.	Taxes @ 35%	2,87,000	1,34,400
H.	Earnings After Tax (EAT)	5,33,000	2,49,600

Additional Information:

No. of equity shares	2,00,000	1,60,000
Dividend payment Ratio (D/P)	20%	30%
Market price per share	₹ 50	₹ 20

Assume that both companies are in the process of negotiating a merger through exchange of Equity shares:

You are required to:

- Decompose the share price of both the companies into EPS & P/E components. Also segregate their EPS figures into Return On Equity (ROE) and Book Value/Intrinsic Value per share components.
- Estimate future EPS growth rates for both the companies.
- Based on expected operating synergies, R Ltd. estimated that the intrinsic value of S Ltd. Equity share would be ₹ 25 per share on its acquisition. You are required to develop a range of justifiable

Equity Share Exchange ratios that can be offered by R Ltd. to the shareholders of S Ltd. Based on your analysis on parts (i) and (ii), would you expect the negotiated terms to be closer to the upper or the lower exchange ratio limits and why?

International Financial management

13. A proposed foreign investment involves creation of a plant with an annual output of 1 million units. The entire production will be exported at a selling price of USD 10 per unit. At the current rate of exchange dollar cost of local production equals to USD 6 per unit. Dollar is expected to decline by 10% or 15%. The change in local cost of production and probability from the expected current level will be as follows:

Decline in value of USD (%)	Reduction in local cost of production (USD/ unit)	Probability
0	-	0.40
10	0.30	0.40
15	0.15 Additional Reduction	0.20

The plant at the current rate of exchange will have a depreciation of USD 1 million annually. Assume local Tax rate as 30%.

You are required to find out:

- Annual Cash Flow After Tax (CFAT) under all the different scenarios of exchange rate.
- Expected value of CFAT assuming no repatriation of profits.
- Viability of the investment proposal assuming an initial investment of USD 25 million on plant and working capital with a required rate of return of 11% on investment and on the basis of CFAT arrived under option (ii). The CFAT will grow @ 3% per annum in perpetuity.

Theoretical Questions

14. "Securities are created by dividing the cash flows associated with underlying securities into two or more new securities". Explain the instruments & its types.

15. Explain the various technique explicitly does not involve transaction costs and can be used to offset the foreign exchange exposure completely or partially.